WESTFIELD CHILDREN'S CENTRE

PART 1
LECTURE 17

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Westfield at a Glance

- Westfield is a large, non-profit pediatric rehabilitation provider serving children and youth (ages 0–17) across the Kitchener–Waterloo and Wellington region

- With a network of **7 clinics** across the region, they support over **17,000 clients each year** through services such as:
  - Speech-language therapy (communication, speech & language development)
  - Occupational therapy (daily living skills, fine motor, sensory needs)
  - Physical therapy (mobility, strength & gross motor development)

Programs Offered at WestField

- **Community-Based rehab services (CBRS)**
  - Care is delivered **in-clinic** (7 locations) or virtually
  - Primarily supports preschool-aged children (0-4 years old)
    - Also referred as the Early Years (EY) program
  - Families access services through scheduled appointments and clinic-based teams

- **School-Based rehab services (SBRS)**
  - Care delivered **in schools**, integrated into the child's learning environment
  - Supports school-aged children (4-17 years old)
  - Clinicians travel to 100+ school sites across the Kitchener-Waterloo region

Industry Landscape

- Families can access rehab services through two pathways:
  - **Publicly Funded Children's Development Centres** (e.g., WesField)
  - **Private clinics** (e.g., Toronto Children's Therapy Centre)
- What this means for families:
  - Public care is broadly accessible and low/no cost, but demand often exceeds capacity → long waitlists and delayed starts
  - Private care can get clients started sooner and offers scheduling flexibility, but costs are high ($120+/hr+) which makes it less universally accessible

Key Trade-off: Public care is affordable but slow to access; private care is faster but cost-prohibitive for many families

Toronto Children's Therapy Centre

Speech-Language Therapy · Occupational Therapy · Group Programs

Speech Language Therapy Fees

SPEECH-LANGUAGE FEES

SLP = speech-language pathologist

CDA or **SLA** = communicative disorders assistant or speech language assistant

** For home / school visits – please add $35 per visit for travel*

SLP Assessment: **$255**

60-minute appointment & summary letter

SLP Treatment 60: **$170**

50-min session + 5-10 min communication with caregiver, update child's records and plan next session

SLP Treatment 45: **$127.50**

35-min session + 5-10 min communication with caregiver, update child's records and plan next session

ORGANIZATIONAL CHALLENGES

Organizational Challenges at WestField

- WestField faces a significant issue of long wait times for service
  - 5,000 patients currently on waitlists
- This is a **systemic issue** in pediatric rehabilitation services
- Wait times for treatment across the province are rising 13-14% year-over-year

Long Waiting List For In-School Speech Therapy Means Some Local Kids Are Falling Behind

By **Pamela Vanmeer** November 10, 2021

Ready to right-size Ontario's pediatric health system

TORONTO, March 23, 2023 – The Children's Health Coalition welcomes the Ontario government's commitments to allocate additional funding and work with frontline pediatric health partners to identify more ways to connect children and youth to the care they need.

Toronto

Wait for core Ontario autism services tops 5 years: advocates

Finding comes from survey conducted by coalition of advocates after families reported long waits for services

Allison Jones · The Canadian Press · Posted: Oct 30, 2025 11:01 AM EDT | Last Updated: October 30, 2025

Why Wait Times Persist: Demand Drivers

- Demand issue (more people needing care)
  - **Rising prevalence / identification** of communication and developmental needs

London · **Video**

Number of London-area kids who need speech help in kindergarten has tripled, report says

The number of kids in Grades 1-8 with complex communication needs has doubled

Kate Dubinski · CBC News · Posted: Jan 08, 2025 5:00 AM EST | Last Updated: January 8

  - Referrals for pediatric speech-language services have **nearly doubled** in the last 3–4 years
  - **Complexity of cases increasing** (more high-needs / longer episodes of care)

Why Wait Times Persist: Supply Drivers

- Supply issue (not enough capacity)
  - **Lack of clinicians to hire** to expand clinical capacity
    - The Canadian government currently expects a **moderate risk of shortage** of clinicians over the next decade
    - Their estimates are relative to the number of currently employed clinicians, of which we already have a significant shortage relative to service demand

Why Wait Times Persist: Process Drivers

- Process issue (capacity exists, but isn't always converted into visits efficiently):
  - **Scheduling complexity** across sites, schools, and virtual delivery can create unused gaps
  - **Cancellations / no-shows** reduce realized capacity
  - **Intake + triage workflow** may route some clients later than necessary or to mismatched service intensity
  - **Variation in practice** (session length, frequency policies, modes of therapy offered) affects throughput

Where Can We Act?

- In publicly funded rehabilitation, the traditional levers are constrained.
  - Demand cannot easily be reduced (turning away children is not an option)
  - Supply cannot quickly expand (hiring clinicians and securing funding takes time)
- This leaves process as the most immediate lever for change– how the system operates day-to-day:
  - E.g., how referrals move through intake
  - E.g., how appointments are scheduled
  - E.g., how service is delivered
- Unlike demand or supply, process can be studied, redesigned, and improved without new resources or policy changes
- Before any redesign is possible, the problems must be made visible — measured, mapped, and understood

WHAT IS HAPPENING AND WHY?

Descriptive + Diagnostic Analytics

Descriptive & Diagnostic Analytics

- Before we can improve a system, we need to understand it. This lecture asks two questions: What is happening? And why do we see these problems occur?

Descriptive

"What happened?"

Example: Waitlist length and wait-time distributions, cancellation rates

Diagnostic

"What factors caused it to happen?"

Example: treatment length variability, eligibility policies

Predictive

"What will happen?"

Example: Time-series forecasting

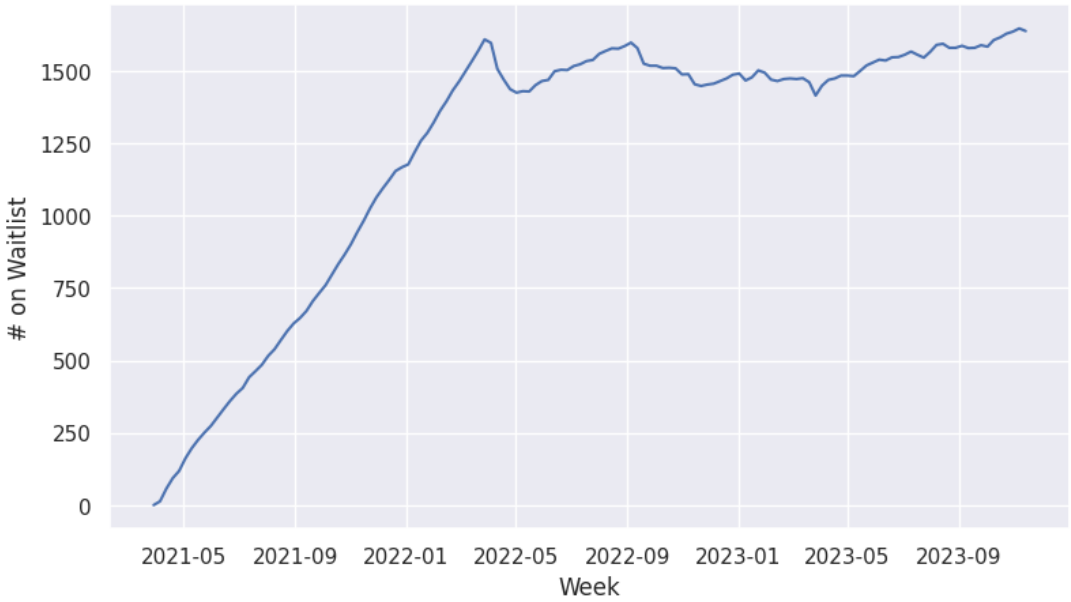
Prescriptive

"What should we do?"

Example: Capacity allocation models & scheduling optimization

What Do Wait Times Look Like?

- Our dataset begins in January 2021 and focuses on the Early Years program (ages 0–4)
- We see that the waitlist grows and then becomes relatively steady from April 2022 – present
  - Post-April 2022 mean=1520.67
- The plateau does not mean the problem is "solved" as some demand is simply exiting the system as clients age out (i.e., become ineligible for early years services)



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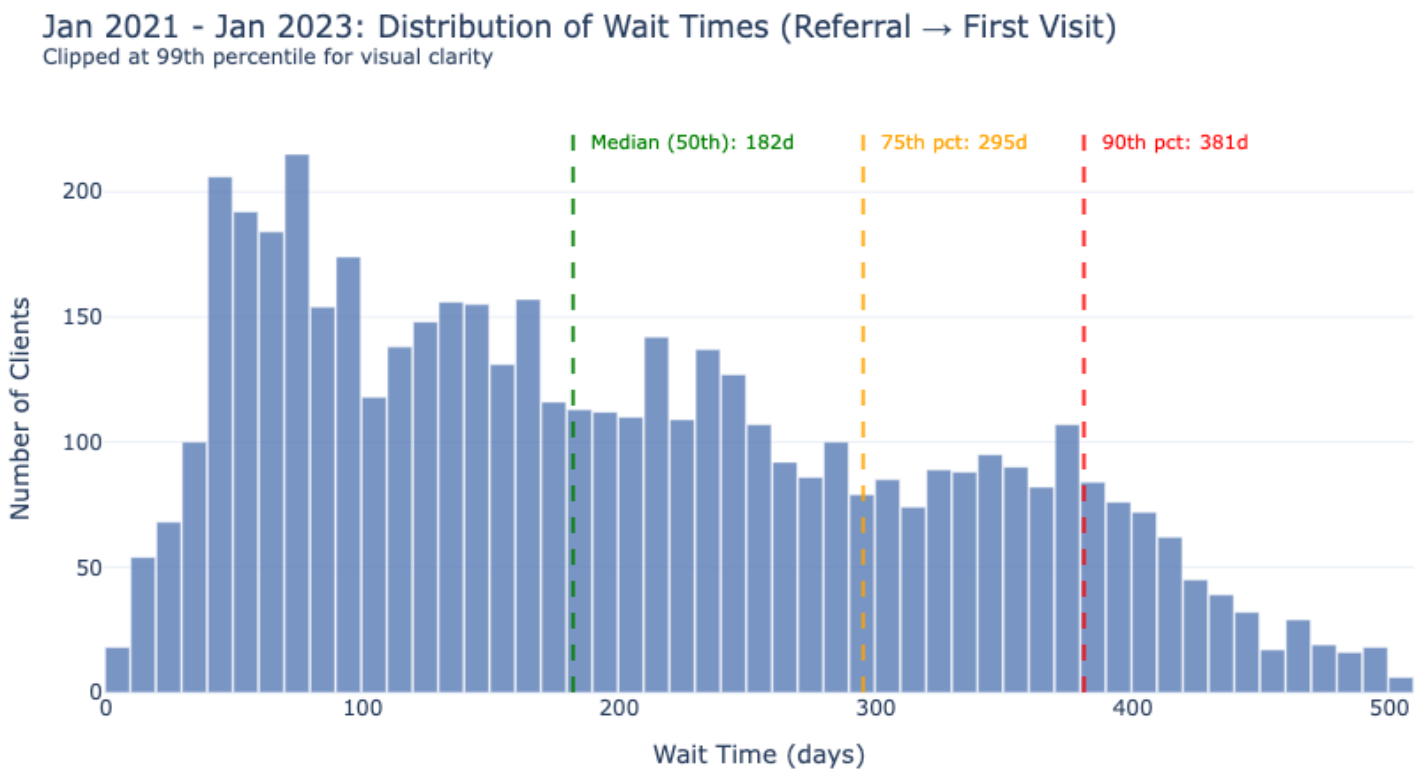
Modeling Wait Times as a Distribution

- We can see that half of all clients **wait over 6 months** before their first appointment
- **A high-risk subgroup (90th percentile) waits over a year**, which is concerning because Early Years eligibility is time-limited
  - Delays push children closer to aging out and can mean missing support during a critical developmental window.
  - Example: a 1 year wait for a child at age 2.5 leaves only ~6 months for treatment time



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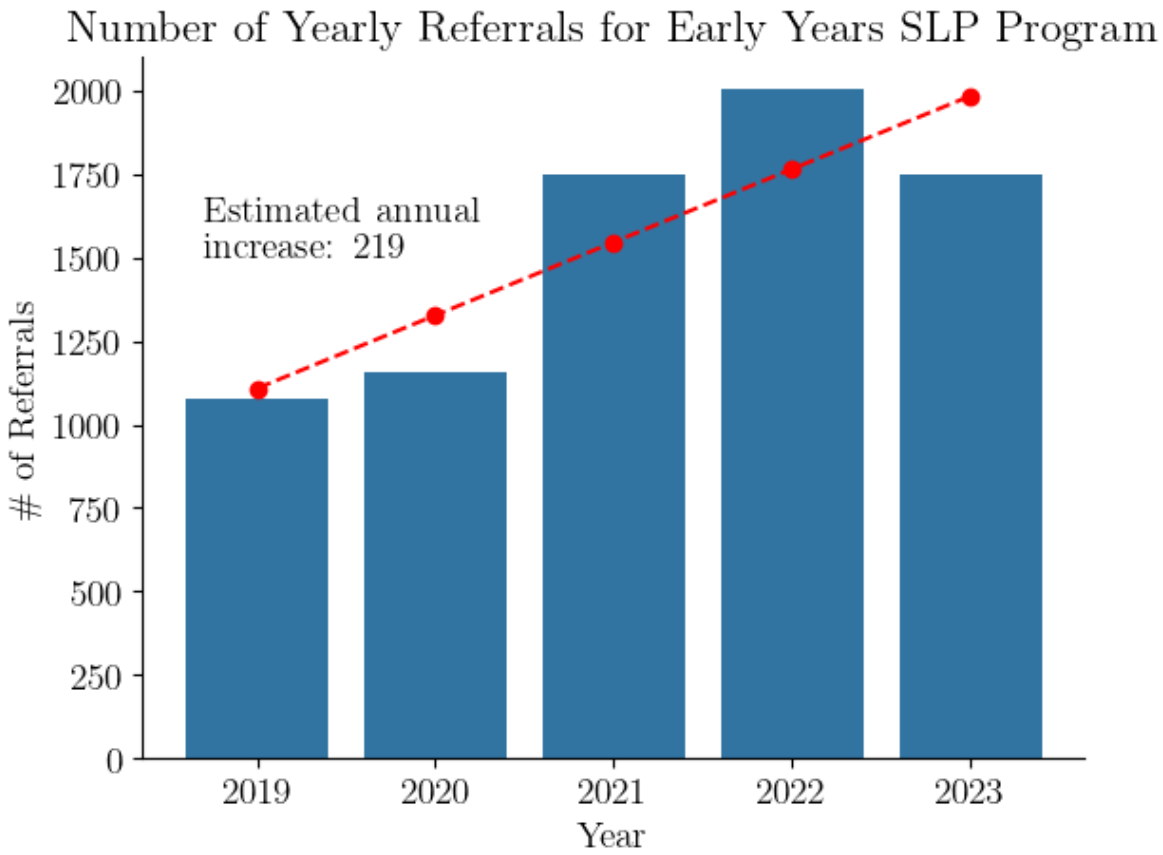
Modeling Wait Time as a Queue

- Queuing theory gives us a way to model these wait times
- The waiting time in the queue is a function of the service time (μ), arrival rates (λ),and servers in multi-server queues
- Increasing arrival rates = increasing wait times if service rates or the number of servers do not increase



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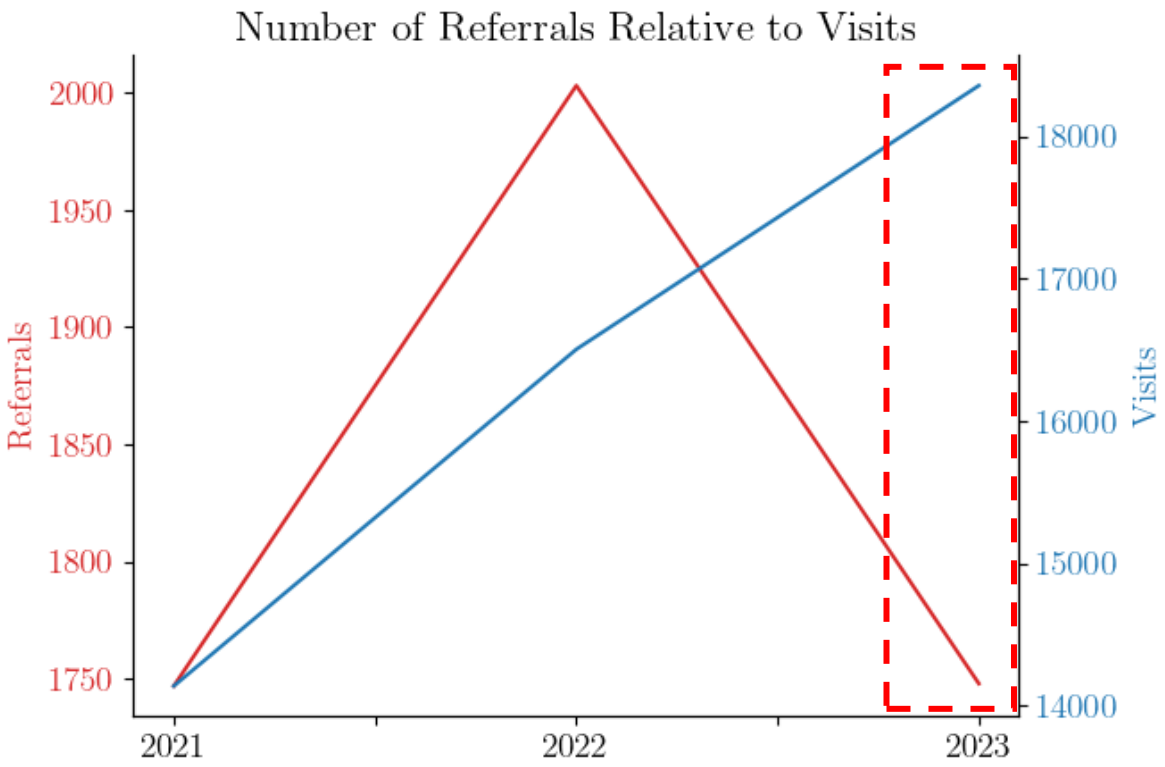
Arrival Rate vs. Service Volume

- Arrival rates have grown ~15.1% YoY
- Meanwhile, the number of appointments delivered per year has grown ~14% YoY
- How does this affect wait times?
  - At best, wait times will remain roughly the same
  - At worst, wait times will grow given the system is already overwhelmed
- Although WestField is delivering more appointments each year, demand is rising at a similar or higher rate, and because there's already a backlog, matching demand growth is not enough to reduce wait times.



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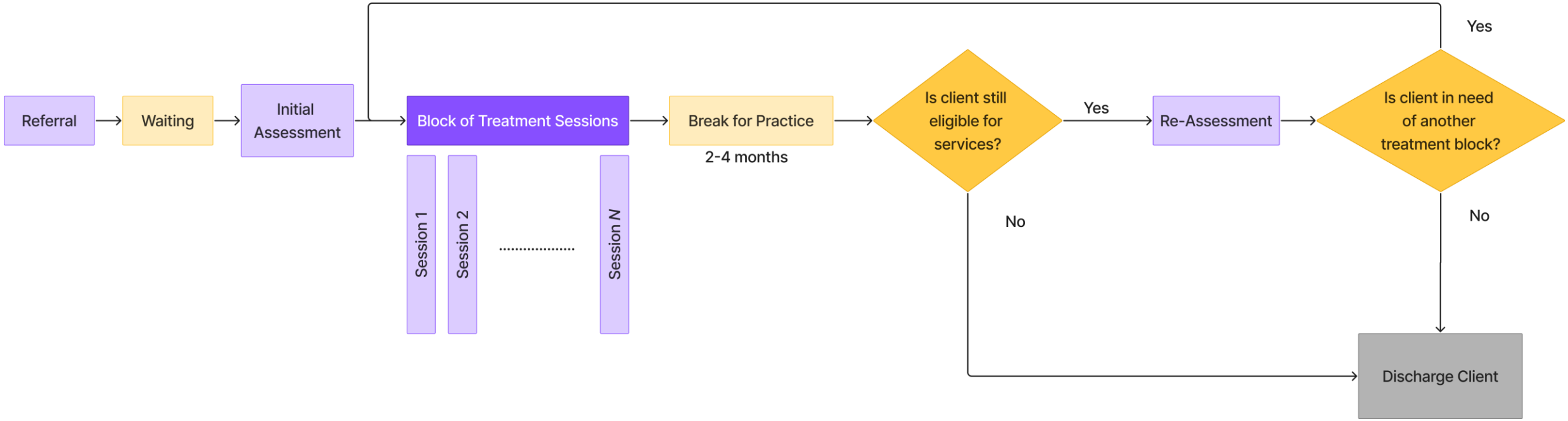
Multi-Appointment Care

- In a classic M/M/1 queue, a “customer” arrives, receives one service interaction, and departs (immediately freeing up capacity)
- Rehabilitation services is multi-appointment: a client receives a **sequence of sessions** (referred to as a treatment block)
- The **Break for Practice** gives families time to practice and generalize skills in daily life so the clinician can reassess later and decide whether to continue, adjust, or discharge



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Service Time and Capacity Allocation in Multi-Appointment Care

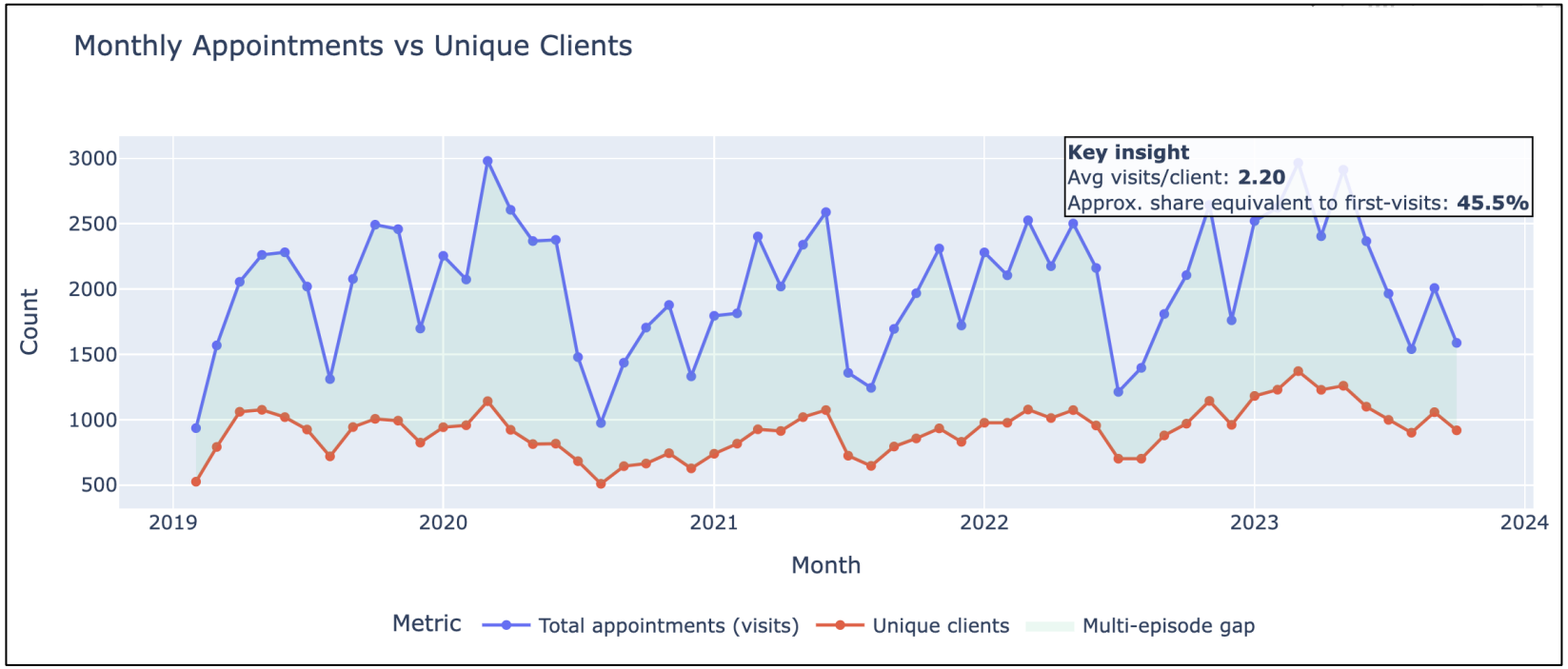
Treatment Block 6-10 sessions	**Break** 2-4 months	**Re-assess**	Discharge or start another block

← - - - - - - - - Full episode (client is still considered "active" until discharged) - - - - - - - - - - - - - - →

- Clients often return from their break and go through additional blocks of therapy
  - "Service time" is therefore based on a client's **entire episode** — with high variability in session count and frequency
  - Active clients have a reserved claim on future capacity
  - Capacity planning must forecast for the full episode load (including clients currently on break) before determining how much room exists to take on new referrals

Appointments ≠ Unique Clients

- Monthly visits are consistently **~2.2 per client on average**
- So even if "appointments delivered" increases, **new clients served (throughput)** may not increase proportionally → waitlists may not shrink.



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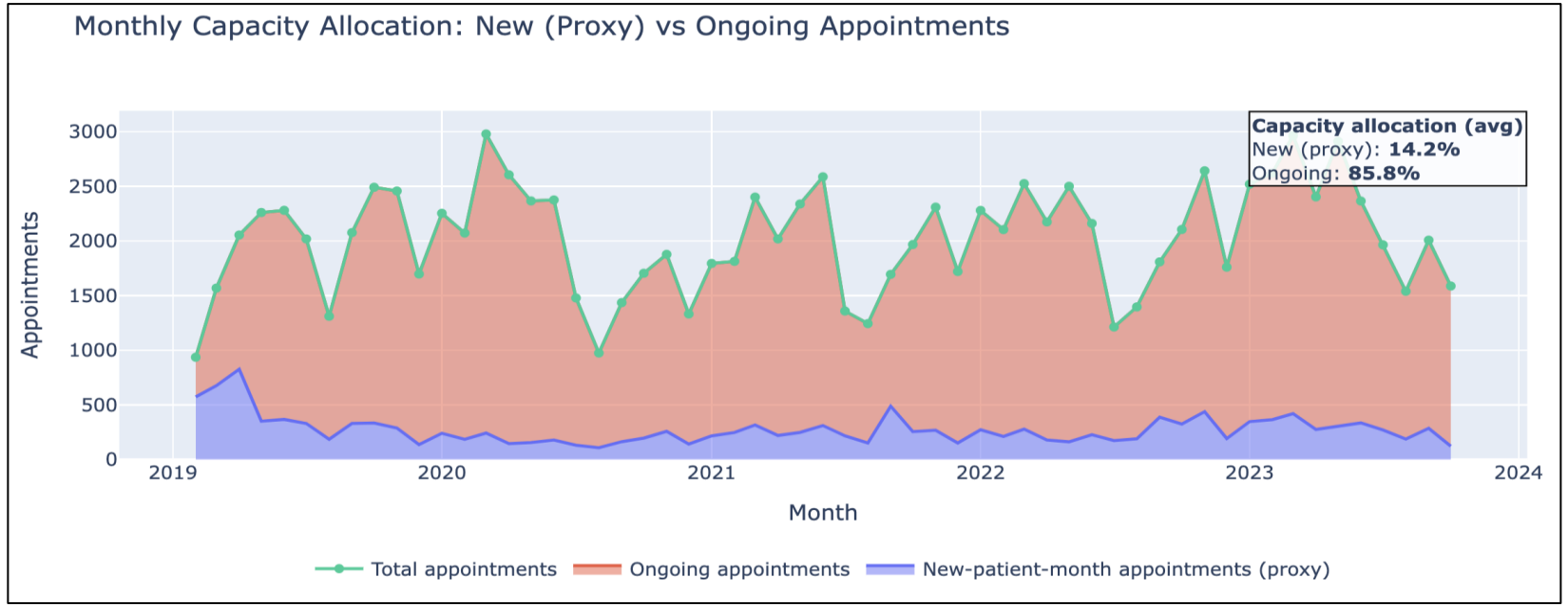
Capacity Allocation: New Starts vs Ongoing Care?

- Most delivered visits are ongoing care
- **Access implication:** total appointment volume can remain high while new-start throughput stays limited
- This is due to the multi-session and episode-based care, so once a client starts, they "reserve" future capacity across weeks/months



Waitlist reduction is driven less by *total visits* and more by the **share of capacity reserved for new starts**

"New start" proxy = client's first appointment month observed in the dataset

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How variable is treatment intensity?

- Since most capacity is tied up in ongoing care, we examine treatment intensity (appointments per client) to quantify how episode length varies and how that variability drives capacity demand
- We see that treatment intensity is **highly variable** (wide-spread)
- A large share of capacity is committed to long-running episodes, limiting room for new starts and **making averages misleading in capacity planning and scheduling**

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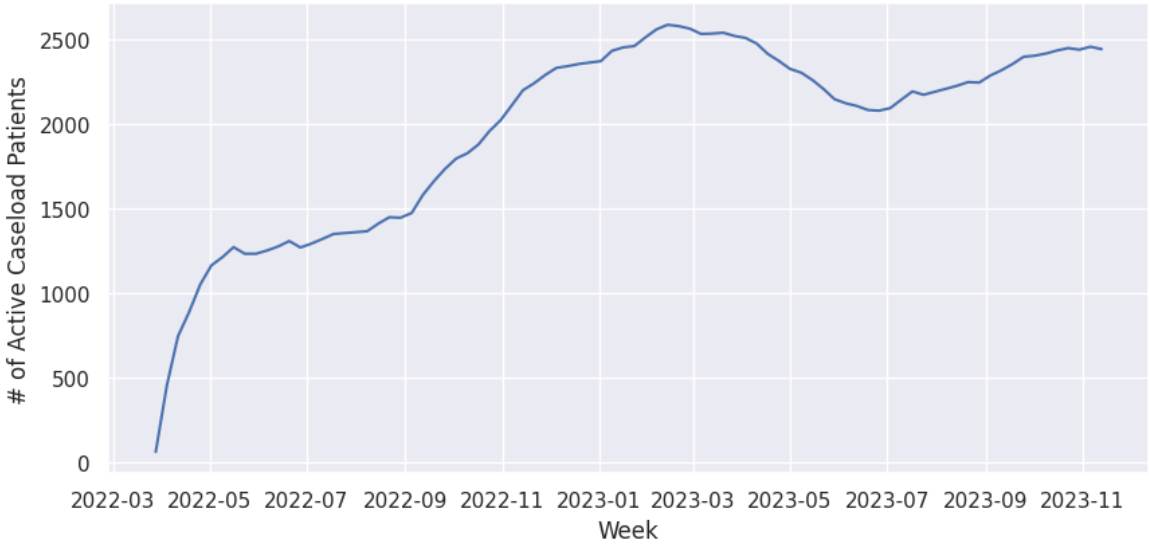
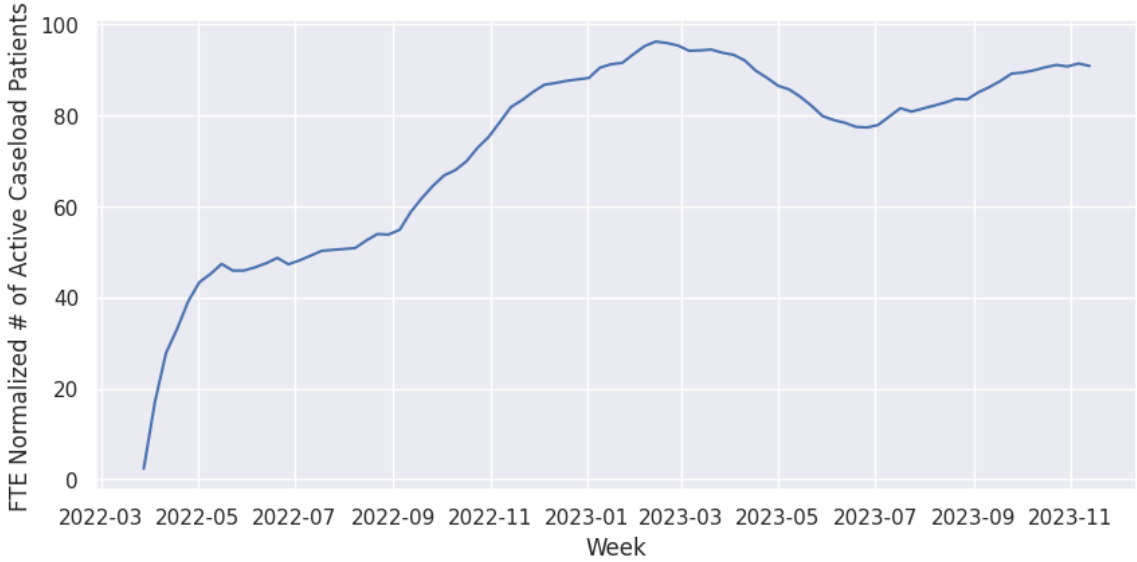
Treatment Capacity

- To accommodate such treatment intensities, we first need to figure out how much capacity we have with the existing clinical resources
- Treatment capacity is a function of the number of clinicians and the number of treatments each clinician can deliver per week
- Full Time Equivalent (FTE) is used to measure the contracted hours for a staff member
- 1 FTE = 20 visits/week (approximately)
- 30 FTEs assigned to the Early Years program
  - 30 * 20 = 600 visits per week

Active Caseload

- Active caseload is composed of patients who have been assessed and are currently receiving treatment
- There have been 2000-2500 patients on active caseload for the past year
- More important than the absolute number is the FTE-adjusted # of active caseload patients
- We can see that on average, each FTE actively manages roughly 80 – 96.25 patients
- Treatment capacity has impacts on the ability to deliver high frequency to patients on active caseload





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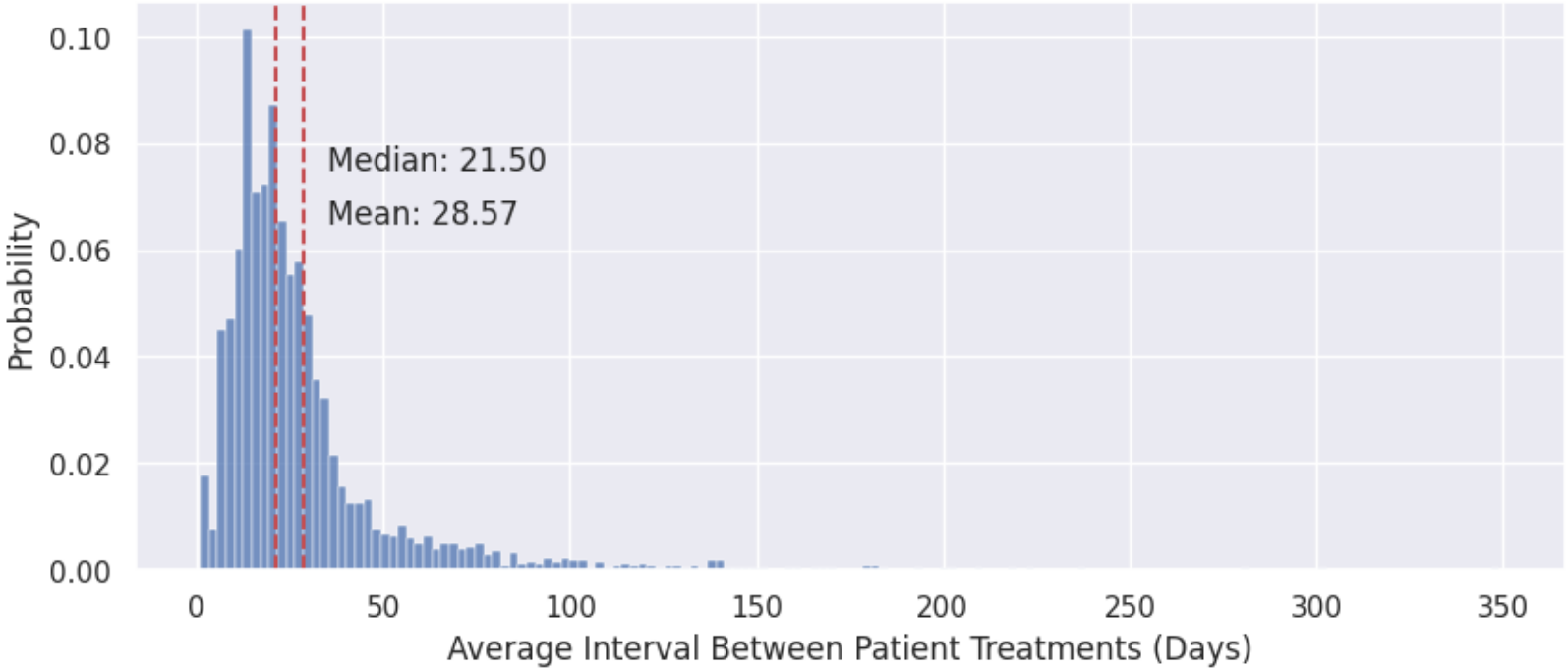
Understanding Active Caseload from a Scheduling Perspective

- Consider that patients are set to receive treatment at some interval of x weeks
- If a clinician is actively managing 80 patients, that means that they can see each patient once every 4 weeks
- If we instead consider that patients receive treatment in a 2-block format (i.e. they receive the first half of their prescribed treatments, take a 2-3 month break to practice, and then receive the other half) they could receive treatment every 2 weeks

Treatment Frequency Patterns

- We see that the average treatment frequency for patients roughly equates to what was estimated from the FTE-normalized active caseload
  - i.e., If each clinician is carrying ~80–96 active clients, the system naturally produces roughly ~3–4 weeks between sessions



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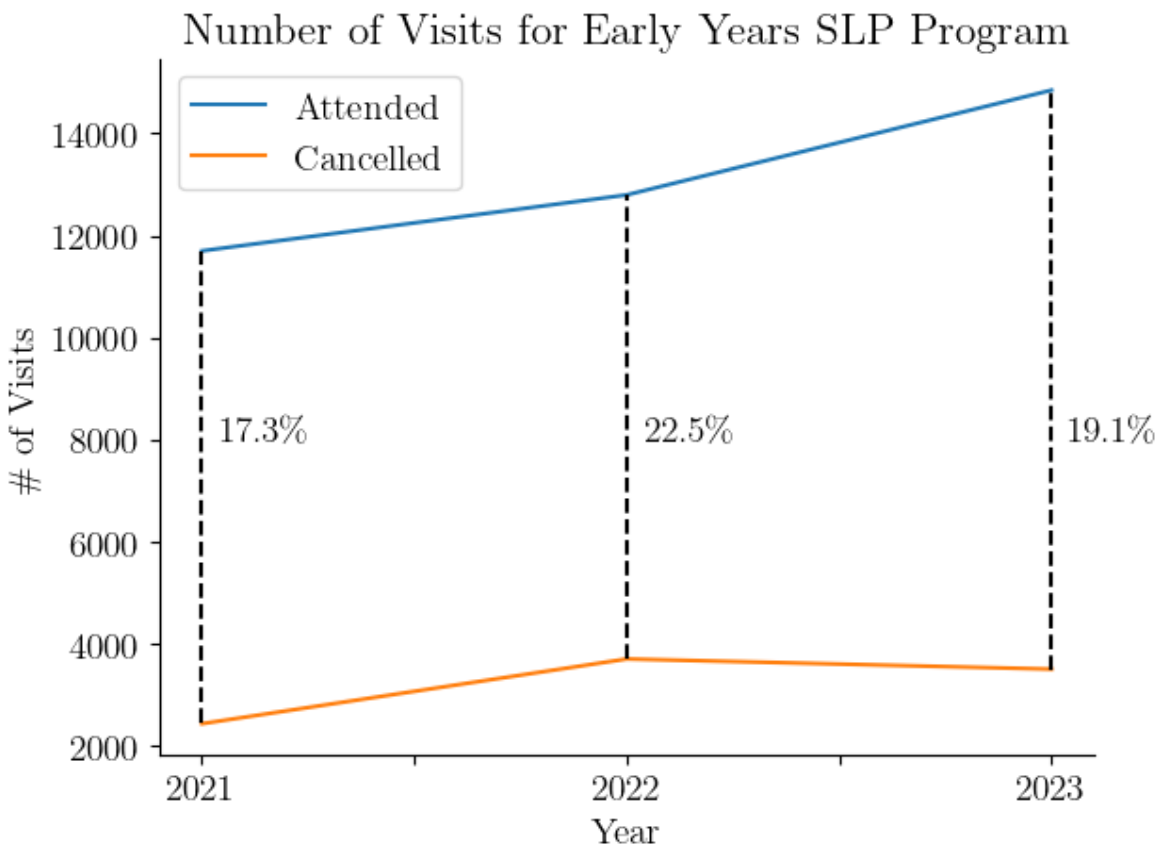
A Hidden Driver of Wait Times: Cancellations

- In many service settings, the number of appointments does not tell the whole story.
- Appointments can be no-showed/cancelled
- In the case of WestField, ~20% of appointments do not occur
  - The annual proportion has been marginally growing over time
  - Reasons?
    - During COVID, it was easier for families to meet their appointment requirements given most people were working remotely and had more flexibility in their schedules + virtual visits were more common



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Not All Missed Appointments Are Equal

No-Shows

~4.2% of appointments

Client misses an appointment with no notice. Slot is 100% lost (impossible to fill last-minute).

Slot recovery: 0% filled

CAPACITY IMPACT: HIGH

Last-Minute Cancel (<24 hours)

~6.6% of appointments

Client cancels within 24 hours.

Slot recovery: 2% filled

CAPACITY IMPACT: HIGH

Advanced Cancel (>24 hours)

~8.1% of appointments

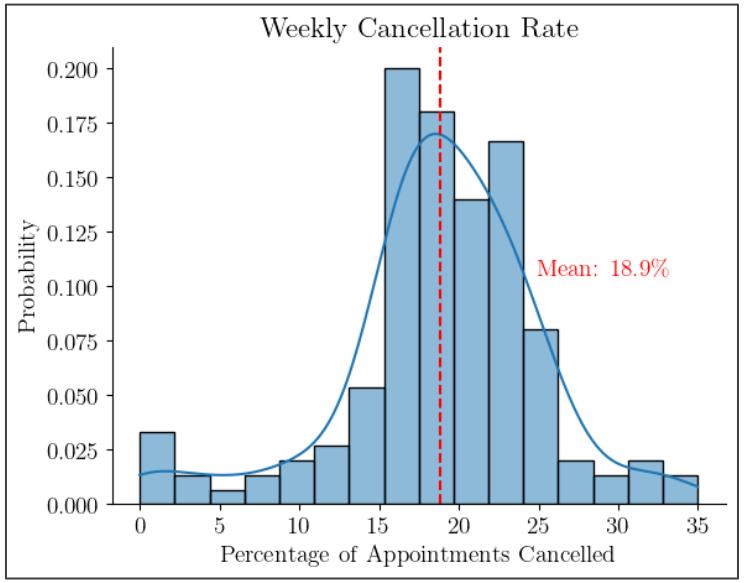
Client provides advance notice. Some recovery is possible. Sometimes results in clinician rebooking client for later in the week.

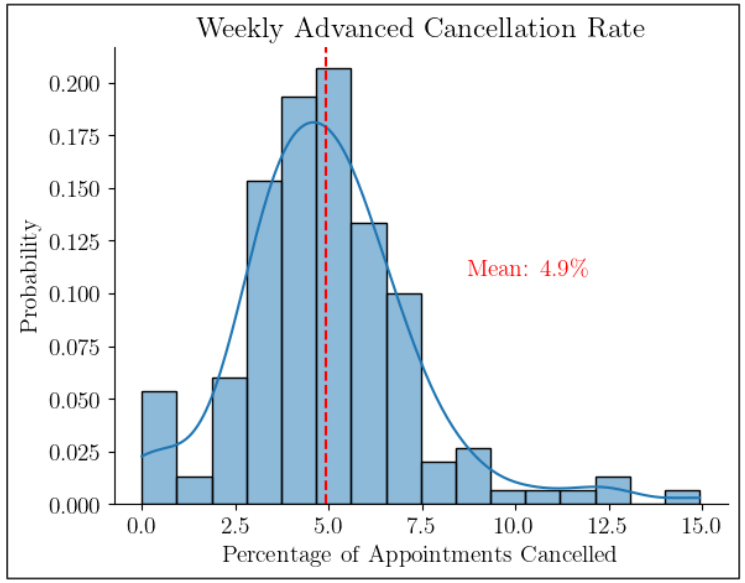
Slot recovery: 7% filled

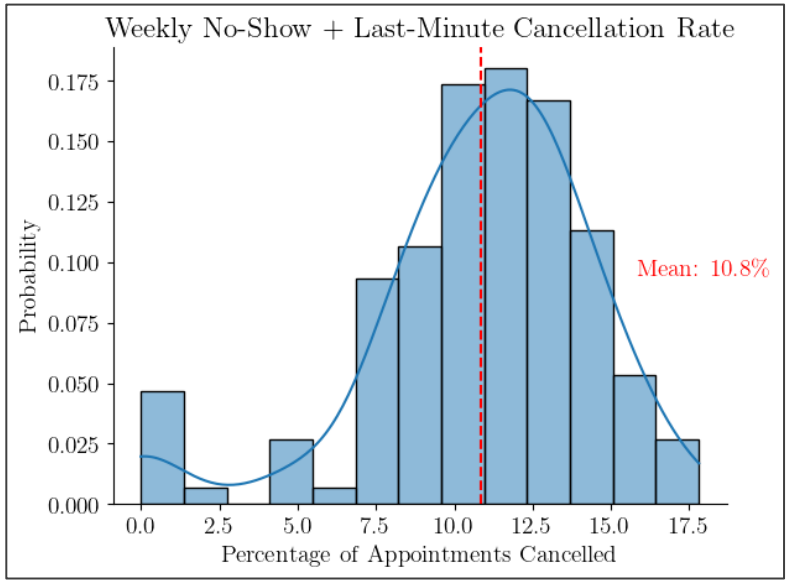
CAPACITY IMPACT: MODERATE

Weekly No-Show/Cancellation Rates

- Advanced cancellations are right-skewed
- No-show and last-minute cancellation rates seem to be left-skewed
- What does this mean?
  - The tail end risk of having significantly more advanced cancellations than the average is higher
  - The tail end risk compared to the average is that no-shows and last-minute cancellations will be lower than the average







Service Delivery Trade-offs

- So far, the data tells us the system is capacity-constrained. That forces a harder question:

When you cannot serve everyone optimally – how do you decide who gets what, and when?

This is not just an operational question but an ethical one. Should we:

Serve everyone a little	vs.	**Server fewer people fully**	*Breadth vs. depth*
First-come, first-served	vs.	**Prioritize highest-need**	*Fairness vs. urgency*
Maximize total visits	vs.	**Maximize new-client throughput**	*Volume vs. access*

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Aging-Out Rush Service Problem

- Recall that the Early Years Program serves children aged 0-4 years old
- We can see that the program eligibility age range in Early Years treatment can lead to a system problem
- There is **limited treatment capacity** (function of: # of clinicians, # appointments per week per clinician)
- Limited window for treatment delivery:
  - E.g. Patient A arrives at age 2.5. Patient A requires 20 treatments. The ability to fit these 20 visits in before Patient A ages out of program eligibility is a function of the frequency at which these appointments can be delivered (weekly=20 weeks, bi-weekly=40 weeks, etc.)
  - This does not even account for the effect of **waiting**

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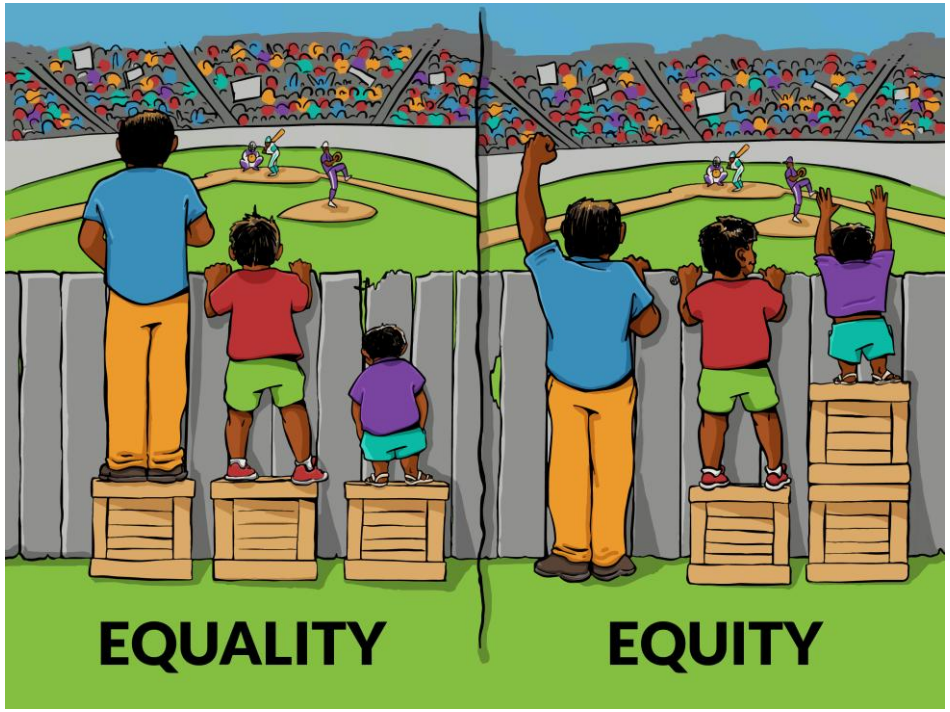
Equality vs. Equity

- Due to demand-capacity inequalities, we must consider trade-offs between equality and equity
- **Equality**: all patients receive the same number of sessions
  - Simple, transparent, and defensible
  - Assumes all clients have homogenous needs
  - May over-serve mild cases, and under-serve severe cases
- **Equity**: all patients receive the necessary treatment amount, determined by their clinical need
  - Best clinical outcomes, ensuring vulnerable patients get adequate care
  - Harder to implement
  - May increase wait times for some while complex cases get more sessions

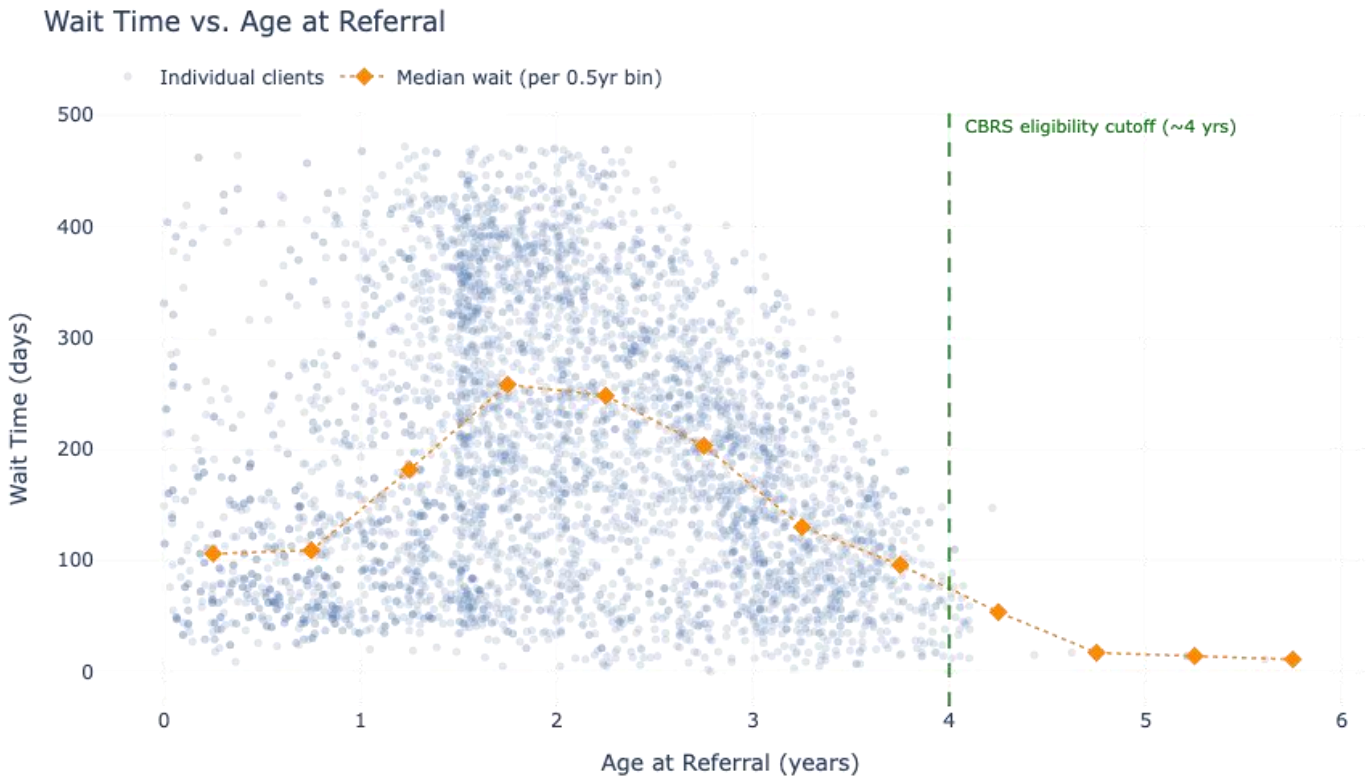


Reality vs. Utopia

- Healthcare is, unfortunately, often a field in which reality makes achieving equitable outcomes extremely difficult, if not impossible, in the short-term
- Oftentimes, factors outside of an organization's control (e.g., government funding, lack of available staff to hire) render it near-impossible to achieve equitable outcomes
- Thus, organizations are forced to assess how they choose to balance these paradigms
  - For example, do we guarantee some treatment to all patients, at the cost of many patients not as much treatment as they would ideally get? How do we choose the thresholds for this?
  - Do we guarantee equitable treatment at the cost of very long wait times?
  - Do we make hard short-term policy choices for the benefit of positive outcomes down the road?
  - How do we set thresholds?

Aging-Out Prioritization Policy in Practice

- The curve shows wait times are highest around 18–24 months, then **decline after age 2**, with the steepest drop after ~3.5 years as children approach the **4-year eligibility limit**
- Children referred at 0–1 years are likely flagged with **high-severity or medically urgent conditions** (e.g., permanent hearing loss or conditions identified at birth). These cases would naturally receive faster prioritization
- The system does appear to respond to aging-out urgency, but only meaningfully close to the cutoff



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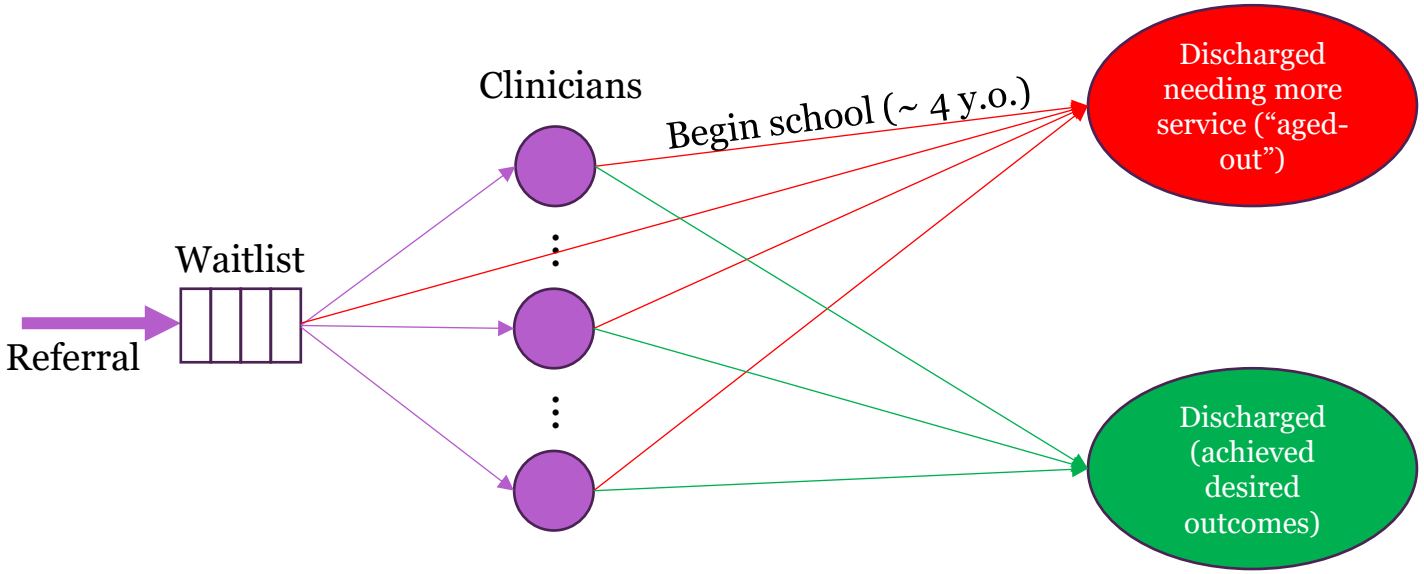
Service Style and Waitlist Mitigators

- Unlike in many service settings, there is a quasi-buffer on the waitlist length
- When patients begin school, they "age-out" of program eligibility and get transferred to the SBRS program waitlist
  - Benefits: prevents the waitlist from growing infinitely
  - Cons: regardless of how much service a patient has received, they will be put back on another waitlist



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Downstream Effects of the Quasi-Buffer and Aging-Out

- For Early Years, the "aging-out" valve limits runaway growth – once children reach school age, they exit the CBRS waitlist
- **But demand doesn't disappear – it shifts:** Those children move into school-based services, where eligibility spans a much broader age range (~4–17 years), so they enter a much larger queue
  - Greater congestion → longer waits, more variability, and higher risk of disrupted or delayed care
- Clinical consequence: When support is delayed, difficulties can compound over time, increasing the likelihood that mild needs progress into more complex, higher-intensity cases

Hypothesis Testing

- A formal statistical method for deciding whether an observed difference between groups is **real** or simply due to **random chance**

Null Hypothesis (H_0)

There is no REAL difference. Any difference seen in the data is due to random variation.

Null Hypothesis (H_1)

There is a real difference between the groups being compared.

- In the WEstField context, we ask:
  - Did appointment volumes meaningfully change (increase) year to year?
  - Did cancellations rate meaningfully change year to year?
- We use a t-test to formally **compare the means** of two distributions and calculate a p-value
  - Using $\alpha=0.05$, if $p < 0.05$ we can reject H0 and conclude the difference is statistically significant

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Appointment Volume

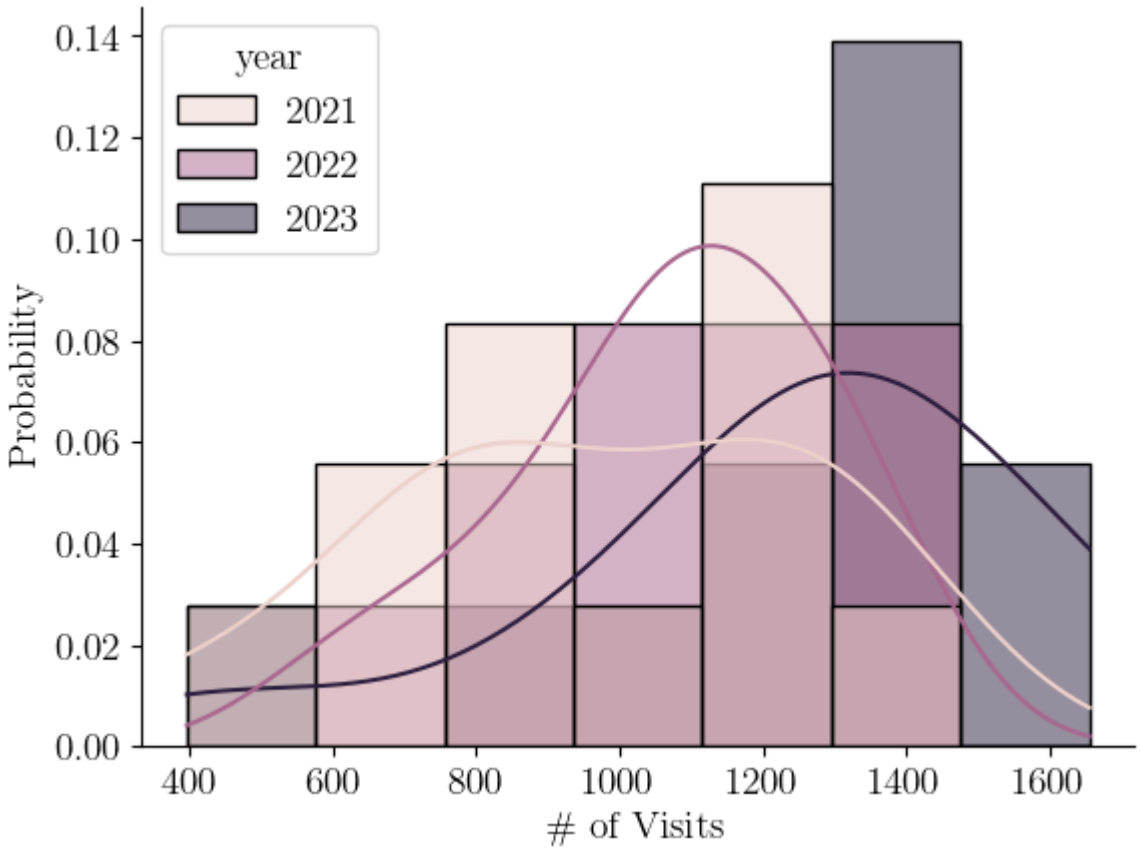
T-test (with unequal variances): distributions of monthly visit volume by year

Years	Mean Diff	T-statistic	P-value
2022 vs. 2021	91.3	0.85	0.41
2023 vs. 2021	262.2	2.02	0.06
2023 vs. 2022	170.9	1.49	0.15

Although average monthly visits grew (+262 from 2021 to 2023), the increase is not statistically significant at $\alpha = 0.05$. The p-value of 0.06 is suggestive but inconclusive.

Conclusion: visit volume has not demonstrably grow fast enough to meet or outpace demand

When two distributions heavily overlap, the t-test cannot confidently tell them apart (resulting in a large p-value)



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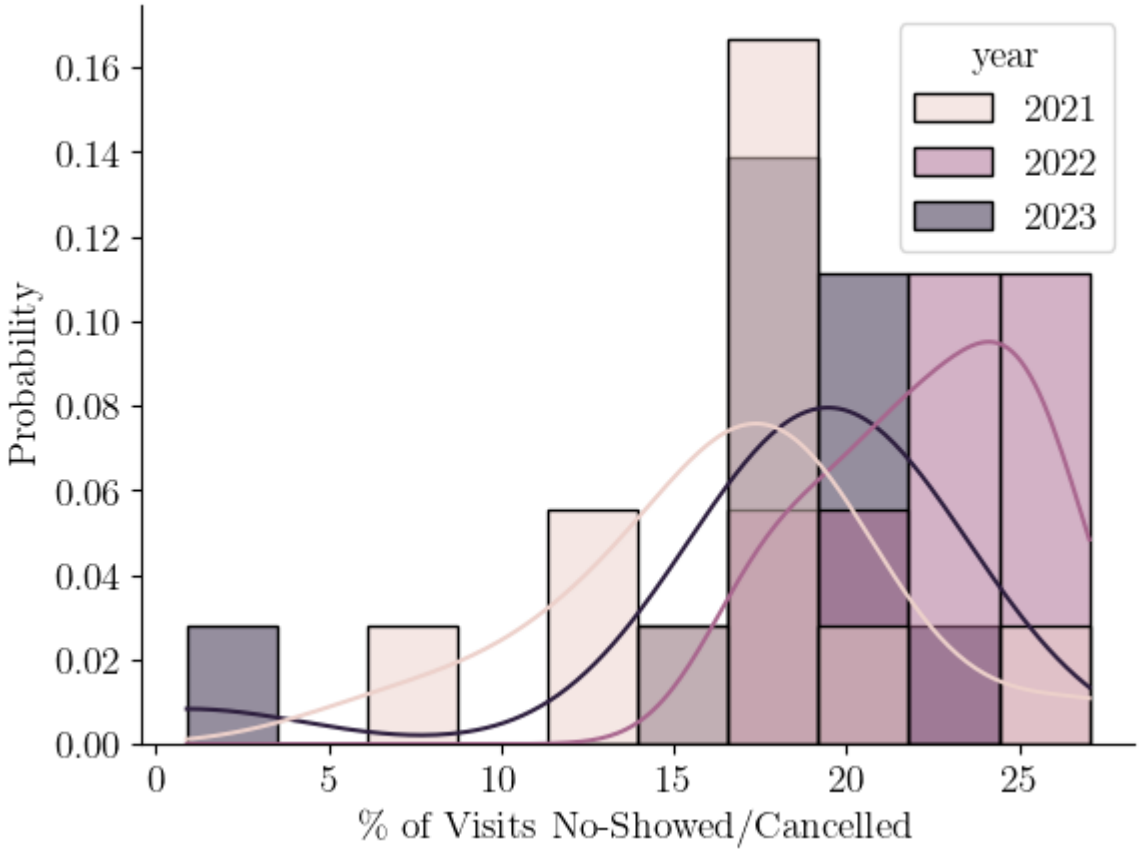
Cancellation Rates

T-test (with unequal variances): distributions of monthly cancellation rate by year

Years	Mean Diff	T-statistic	P-value
2022 vs. 2021	5.6	3.36	<0.001
2023 vs. 2021	1.2	0.57	0.57
2023 vs. 2022	-4.3	-2.32	0.03

We see that the weekly cancellation rates year-to-year have changed significantly, however, there is not a clear trend

- Increased from 2021-2022 (possibly due to return to in-person schedules post-COVID) but decreased from 2022-2023
- There is a reason for this reduction which will be explained in a later lecture



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Part 1: Main Takeaways

1. **WestField faces an access management issue:** 5,000 patients on waitlist, median wait of 182 days, with 10% waiting over a year before starting therapy.

2. **Arrival rates outpace service volume growth**. Because a backlog already exists, simply matching demand growth is not enough to reduce wait times.

3. Multi-appointment, episode-based care means active clients "reserve" future capacity. **Majority of all visits go to ongoing care**, limiting new-start throughput.

4. **Treatment intensity is highly variable**, making averages misleading for capacity planning and locking capacity into long-running episodes.

5. ~20% of scheduled appointments go uncompleted due to **cancellations/no-shows**. This is a hidden but potentially recoverable source of lost capacity.

6. The system faces **unavoidable trade-offs**, and the aging-out mechanism simply shifts unmet demand downstream into an even more congested school-based queue.